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SECTION 02

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1.0 Problem Statement

Many Mandarin learners struggle to correctly write and recall Chinese characters. They frequently forget the stroke order and lose interest as a result of traditional memorisation techniques that are repetitive and boring. This slows down their learning and frustrates them. Therefore, there is a need for an interactive and interesting learning system that can lead users step-by-step, provide them immediate feedback on their writing accuracy and eventually improve their ability to remember characters.

2.0 AI Solution

Rydit: Learn Mandarin is an AI-driven mobile application that transforms traditional Mandarin writing practice into an interactive and personalized learning experience. The app uses artificial intelligence to **analyse user handwriting** in real time, by recognising the stroke order, directions and overall accuracy. The app uses AI handwriting recognition to compare the user's input with the correct stroke sequences and immediately provides visual feedback, guiding the learner to improve precision and consistency in writing Chinese characters.

Apart from handwriting recognition, the AI also powers the app's **adaptive learning engine**. It tracks each user's performance every time they answer a question and dynamically adjusts lesson difficulty based on their strengths and weaknesses. Using the Leitner spaced repetition method [1], Rydit ensures that characters and words the user struggles with will reappear more frequently, while mastered ones are reviewed once in a while to strengthen memory retention. The lessons are challenging but rewarding to engage the users in an immersive learning experience, while learning how to write in mandarin painlessly.

Additionally, Rydit gamifies the entire learning process with progressive levels, rewards and interactive challenges to sustain motivation. The AI monitors engagement patterns and suggests personalised practice sessions to prevent boredom or burnout. By combining adaptive AI feedback, smart repetition and game-based learning, Rydit offers a fun, efficient, and scientifically grounded way for learners to master Mandarin writing and vocabulary.

3.0 Goal of AI Solution

Rydit: Learn Mandarin aims to give learners a more efficient, customized and entertaining method of learning Chinese character writing. By teaching users step-by-step and providing real-time feedback on stroke order, direction and structure, the AI seeks to increase writing accuracy [2]. Additionally, AI adjusts the complexity of lessons according to user performance, which boosts motivation and learning at a personal pace [3]. Through customized guidance and on-going assessment, research indicates that personalized AI systems can improve retention, fortify self-regulated learning habits and maintain learners' increased engagement [4][5]. By using interactive challenges, adaptive learning and real-time feedback, the AI seeks to help users write more precisely, retain characters longer and maintain the motivation throughout their Mandarin learning process.

4.0 Process of Empathize in Design Thinking

Empathy Map

SAYS ● "How to write the characters correctly? " ● "I can recognize the words but always forgot how to write." ● "I always forget the stroke order." ● "It's so hard to learn Mandarin."	**THINKS** ● "I wish there was a fun way to learn!" ● "I want to learn at my own pace." ● "It would be great if there is an app to help me in learning." ● "Traditional memorization is boring."
DOES ● Find a tuition teacher. ● Watch YouTube tutorials on writing characters. ● Practice on a paper or tablet. ● Seek help from friends. ● Learn from app like Duolingo	**FEELS** ● Frustrated when forgetting how to write. ● Confused about stroke order. ● Excited when recognizing or writing correctly. ● Satisfied when seeing progress or getting feedback
PAINS ● Forgetting too fast and lack of motivation to practice. ● Hard to memorize characters without writing repeatedly. ● Forgetting old characters easily. ● Boring if learning alone.	**GAINS** ● A system that help them to remember in long term ● Step-by-step writing animations for better understanding. ● AI feedback on stroke order and accuracy. ● Fun and interactive experience that keeps them motivated.

5.0 Process of Define in Design Thinking

User:

- An individual who is learning Mandarin language.

Need:

- The user needs a way to remember and recall Chinese characters. The user forgets learned characters fast.
- The user needs a way to practice writing correctly and confidently. The user has identified a pain point associated with the challenge of writing the characters accurately and remembering the stroke order.
- The user expressed a demand for an interesting way of learning the language. The users think that learning Chinese can be boring.
- The users think that they would be more motivated to learn with others but at the same time learning at their own pace.

Insights:

- Retention. Spaced repetition is vital to ensure that learned words are not forgotten.
- Writing Accuracy. Writing practice is also important to ensure the characters are written correctly.
- Motivation. Forgetting characters causes frustration and reduces motivation.
- Engagement. Users would enjoy learning in a gamified way and learning in an interactive way.

6.0 Knowledge Representation

6.1 Character Revision Frequency

Defining repetition frequency of a certain Mandarin character to appear as a question based on the mastery of the particular character.

Rule:

$IF\ user\_answer\_correctly\ =\ false$
$\quad OR\ (error\_rate\ >\ 0.6\ AND\ lessons\_since\_last\_revision\ >=\ short\_interval)$
$\quad OR\ (error\_rate\ >\ 0.3\ AND\ error\_rate\ <=\ 0.6$
$\quad\quad AND\ lessons\_since\_last\_revision\ >=\ medium\_interval)$
$\quad OR\ (error\_rate\ <=\ 0.3\ AND\ lessons\_since\_last\_revision\ >=\ long\_interval)$
$THEN\ show\_character\_question\ =\ true$

Explanations:

The system determines the appearance frequency of questions determined by user performance and time interval since the last exercise done on the particular Mandarin character. There are several conditions of determining whether a question repeat for practice:

- Character that user got wrong in one learning session is repeated immediately
- Character with high error rate (> 0.6) is repeated in a short period of time
- Character with medium error rate (> 0.3) is repeated in a medium period of time
- Character with error rate <= 0.3 is repeated occasionally over a long period of time

This rule ensures that revision frequency of learned Mandarin characters is adjusted accordingly based on error rate and spacing to strengthen memory efficiently.

Predicate Logic/First-Order Logic:

$\forall u\ \forall c\ \forall e\ \forall n\ ($
$\quad User(u) \land Character(c) \land ErrorRate(u, c, e) \land LessonsSinceRevision(u, c, n) \land$
$\quad (\neg UserAnsweredCorrectly(u, c)$
$\quad \lor (e\ >\ 0.6 \land n \geq ShortInterval)$
$\quad \lor (0.3\ <\ e \leq 0.6 \land n \geq MediumInterval)$
$\quad \lor (e \leq 0.3 \land n \geq LongInterval))$
$\quad \rightarrow ShowForRevision(u, c)$
$)$

For any user *u* and character *c*, the character is shown for revision if **any** of the following conditions are true:

1. The user answered it incorrectly in the last session.
2. The user has a high error rate and the short interval has passed.
3. The user has a medium error rate and the medium interval has passed.
4. The user has a low error rate and the long interval has passed.

6.2 Handwriting Recognition

Predicting correctness of Mandarin character writing using stroke order, direction and similarity rules.

Rule:
$IF\ user\_stroke\_order \ = \ correct\_order$
$\quad AND\ user\_stroke\_direction \ = \ correct\_direction$
$\quad AND\ user\_stroke\_similarity \ >= \ threshold$
$THEN\ handwriting\_recognition \ = \ correct$

Explanation:
The "Rydit: Learn Mandarin" system reviews user's handwriting Chinese characters in real time to check whether each character is written correctly. The system examines 3 main features of the handwriting:

1. **Stroke Order**: Rydit checks whether the user writes the strokes in the same order as the standard character.
2. **Stroke Direction**: It analyzes the direction of each stroke to determine it matches the correct writing method.
3. **Stroke Shape and Accuracy**: The system compares the user's stroke shapes to the standard version. If the shapes are close enough within set tolerance levels, the stroke is considered correct.

When all 3 conditions are met, Rydit predicts that the user has written the character correctly. The app then provides immediate positive feedback, helping the user build confidence and develop consistent writing habits. These predictions are based on clear rules and data analysis, ensuring users receive reliable guidance throughout their Mandarin handwriting practice.

Predicate Logic/First-Order Logic:
$$\forall u\ \forall c\ \forall i\ \ (CorrectOrder(u,\ c,\ i) \land CorrectDirection(u,\ c,\ i) \land Similar(u,\ c,\ i) \rightarrow PredictCorrect(u,\ c,\ i))$$

For every user *u*, character *c* and stroke *i*, if the user *u* writes stroke *i* of character *c* in the correct order and the stroke direction is correct and the stroke is similar with standard version, then the system predicts the stroke is correct.

6.3 New Word Introduction

New characters shall be introduced from time to time to ensure progression. The condition to introduce a new word and what word to be introduced has to be determined.

Condition to introduce a new word:

Rule:

$IF\ user\_accuracy\_in\_last\_5\_attempts \geq 0.9\ THEN\ introduce\_new\_word$

Explanation:

The system introduces a new character to the user once they have demonstrated mastery of a current character by achieving a 90% accuracy rate or higher in their last 5 attempts. This indicates that the user has a low error rate and is ready to move on to a new character for further learning.

Predicate Logic/First-Order Logic:

$$\forall u \forall c\ \{User(u) \land Character(c) \rightarrow$$
$$[\forall i\ (i \in 1, 2, 3, 4, 5) \land Attempt(u, c, i, accuracy) \rightarrow (accuracy \geq 90)$$
$$\rightarrow NewWordIntroduced(u, c)]\}$$

For every user *u* and character *c*, if the user has achieved 90% accuracy ($accuracy \geq 90$) on their last 5 attempts *i*, then the system will introduce a new word (character *c*) to user *u*.

Word to be introduced:

Rule:

$IF\ character\_not\_mastered\ THEN\ next\_word\_to\_be\_introduced$

Explanation:

The system maintains a list of characters that are sorted by their difficulty. The difficulty is based on factors such as relevance, usage frequency, and radical complexity. Characters that have not been mastered by the user and are available for learning will be selected and introduced to the user in the next learning session.

Predicate Logic/First-Order Logic:

$$\forall u\ \{User(u) \rightarrow \exists c\ [Character(c) \land \neg Mastered(u, c) \land AvailableCharacter(c)]$$
$$\rightarrow NextWord(u, c)\}$$

For every user *u*, if there exists a character *c* that has not been mastered (*¬Mastered(u, c)*) and is available for learning (*AvailableCharacter(c)*), it will be selected as the next word to introduce.

6.4 Motivation & Reward Mechanism

Reward to engage the users in an immersive learning experience by providing the rewards and points that motivate consistent learning and reinforcement.

Rule:

$$IF\ user\_progress\_increases\ OR\ consistent\_practice = true$$
$$OR\ difficult\_question\_solved = true\ OR\ high\_accuracy\_in\_revision = true$$
$$THEN\ reward\_points\_increases\ AND\ positive\_feedback\_triggered$$

Explanation:

The system encourages the learners by giving reward points and providing positive feedback whenever they perform good learning behaviours. For example, this includes making progress while studying, practising consistently, correctly answering difficult questions, or completing a set of problems with high accuracy. As the reward points increase, the system provides motivating feedback to the user. Various achievements can be unlocked and badges are awarded when users reach certain targets, keeping them engaged and immersed in the learning process.

Predicate Logic/First-Order Logic:

$$\forall u\, \forall c\ Progress(u, c) \lor ConsistentPractice(u) \lor SolvedDifficult(u, c) \lor HighAccuracy(u, c)$$
$$\rightarrow IncreaseRewardPoints(u) \land TriggerPositiveFeedback(u)$$

For every user *u* and character *c*, if the user shows progress, practises consistently, solves a difficult question, or achieves high accuracy during studying and revision, then the system will increase the user's reward points and provide positive feedback to the respective user.

6.5 Adaptive AI Feedback

Adaptive AI Feedback provides personalised guidance to learners by analysing their writing performance in real time. The system evaluates stroke correctness, accuracy patterns and user progress to generate appropriate feedback. This ensures that each learner receives helpful hints, corrections or motivational messages tailored to their current needs, improving the learning experience and making Mandarin writing practice more effective.

Rule:

IF stroke_error_detected=true

> *OR repeated_mistakes_on_character=true OR low_accuracy_in_recent_attempts=true*
> *OR performance_improving=true*

THEN provide_personalised_feedback AND adjust_guidance_level

Explanation:

The system delivers different types of personalised feedback depending on the user's writing behaviour. When the user makes a stroke error, such as writing a stroke in the wrong direction or order, the AI gives immediate micro-feedback to correct the mistake. If the user repeatedly struggles with the same character, the system offers more detailed corrective guidance and suggests targeted practice.

When recent accuracy drops, the system provides supportive hints and simplified explanations to help the learner recover. However, if the user shows improvement, such as making fewer mistakes or achieving more accurate strokes, the system triggers encouraging feedback to reinforce progress.

This personalised feedback mechanism adapts to the learner's performance level, ensuring they stay motivated while receiving the right type of guidance at the right time.

Predicate Logic/First-Order Logic:

$$\forall u\ \forall c\ [(StrokeError(u,c) \lor RepeatedMistake(u,c) \lor LowAccuracy(u,c) \lor Improving(u,c)) \\ \rightarrow (ProvideFeedback(u,c) \land AdjustGuidance(u))]$$

For every user *u* and character *c*, if the user makes a stroke error, repeatedly struggles with the same character, shows low accuracy, or demonstrates improvement, the system will provide personalised feedback and adjust the guidance level accordingly.

7.0 Knowledge Representation Involved to Achieve Goals

The major goal of **Rydit: Learn Mandarin** is to provide learners with an interactive, adaptive and efficient method to learn Chinese characters. In this AI system, knowledge representation involves the use of **logical representations** which are propositional logic and predicate logic to capture user performance, handwriting accuracy, revision history and engagement. These representations allow the AI to make informed decisions about which characters to include in lesson, when to provide feedback and how to introduce new content, ensuring optimal learning outcomes for each individual user.

The system uses **character revision frequency rules** to determine how often each character should reappear for practice. Characters that are recently incorrect, have high error rates or are partially mastered are shown more frequently, while well-mastered characters are reviewed once in a while only. The **Leitner-inspired spaced repetition** strategy is consistently applied to strengthen memory retention without overwhelming the learner.

Handwriting recognition is another critical knowledge representation component. The system models each stroke of a character and predicts whether the user has written the stroke correctly, then provides real-time feedback on stroke order, direction, and overall accuracy. This allows the system to guide learners step-by-step in writing characters correctly, building skill and confidence.

New word introduction is represented with rules that track user mastery over current characters. By analyzing the user’s accuracy, the system introduces new characters once sufficient mastery is demonstrated. Only characters not yet mastered and available for learning are selected as the next word. This knowledge representation ensures progressive and personalized learning.

Additionally, Rydit incorporates **motivation and reward mechanisms** to sustain engagement. User actions, progress and achievements are captured as predicates, allowing the AI to assign points, badges or unlock challenges. These rules ensure that learners are consistently motivated and remain immersed in the learning process.

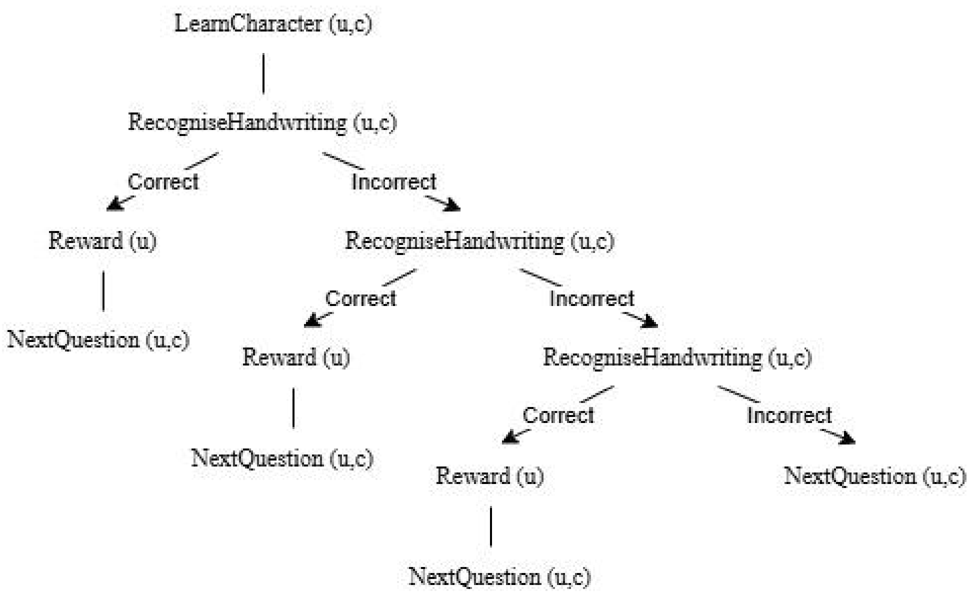
By integrating **adaptive AI feedback, dynamic spaced repetition, handwriting recognition, and motivational rules**, the Rydit system represents a **knowledge-driven and personalised approach** to Mandarin learning. The predicates and rules capture relationships and conditions essential for decision-making, allowing the AI to deliver content, feedback and progression in a way that strengthens retention, improves writing accuracy and maintains learner engagement.

8.0 State Space Diagram

8.1 Introduction

The state space diagram represents the states and actions involved when the user is learning a character, from the selection of the character to be learned, until the character's learning process is complete.

8.2 State Space Diagram



8.3 Description

The characters are labelled as follows for easier differentiation of character status.
Characters Labelling

- "Not Learned": The user has not been introduced to the character
- "Learning": The user is in the process of learning the character and has not yet mastered it
- "Mastered": The user has mastered the character with at least 90% accuracy in the last 5 attempts

State	**Action**
LearnCharacter Determine whether to	At the beginning of the process, a character will be selected to be learned in this cycle.

introduce a new character or revise an already learned character.	The selected character may either be a new character that the user is learning or a newly introduced character. The character to be selected will depend on the number of unmastered characters and the accuracy of the learned characters. First, the system will determine whether a new character should be introduced. If the number of characters labelled as "Learning" is less than the threshold of 5, a new character labelled as "Not Learned" will be introduced. If the number of characters labelled as "Learning" is greater than or equal to the threshold of 5, a character labeled as "Learning" or "Mastered" will be presented. The selected character will be determined based on the mastery level of that character and the interval of revision.
RecogniseHandwriting This state evaluates the user's handwritten Mandarin characters to determine correctness.	In this state, each attempt is assessed based on stroke order, stroke direction and overall similarity to the target character. Every attempt, whether correct or incorrect, updates the character's success and error rates. Repeated incorrect attempts increase the error rate. Once the error rate exceeds a defined threshold, the character will appear more frequently in future exercises to reinforce mastery. The user also receives AI-generated feedback highlighting whether and how their handwriting and performance can be improved.
Reward This state rewards the user if good learning behavior is detected.	There are several situations in which users can be rewarded, including maintaining a continuous learning streak over consecutive days and showing a progress in learning. In addition, users will also be rewarded for solving difficult problems (where the overall error rate of the word is high) or achieving high accuracy in revision questions. When a user is close to reaching a reward state, the system will provide positive feedback to encourage continued engagement. This state motivates users to keep learning until they reach a certain proficiency level and become more skilled in the language.

Goal: NextQuestion

- Which denotes the conclusion of the learning cycle for a chosen character c by user u is the objective state of this state space problem. Once the system has fully assessed the user's engagement with the character through handwriting recognition and where applicable, rewards are provided.

- When the system reaches the NextQuestion state, it has acquired enough data regarding the user's performance, including accuracy, error rate and learning progress. Based on the user's learning history and current mastery level, the system then moves on to presenting the next suitable character or revision work.
- In order to retain user involvement and reinforce effective learning behavior, this goal state guarantees that the learning process is continuous and flexible, enabling the system to dynamically choose the next character.

Path Cost

The minimum path cost shows the least number of checks needed to validate a character. In this case, the user learns the character and gets it right on the first handwriting attempt. Because the answer is correct, the system gives a reward and then moves on to the next question.

The maximum path cost shows the greatest number of checks needed before the system moves on. Here, the user learns the character but fails three consecutive handwriting recognition attempts. Since all checks are incorrect, the system skips the reward and moves directly to the next question.

Solution

A valid solution path is any path from the initial state to the goal state, NextQuestion*(u,c)*.

- **Correct on first attempt**
  LearnCharacter(u,c) →RecogniseHandwriting*(u,c)* = correct → *Reward(u)* →*NextQuestion(u, c)*
- **Correct on second attempt**
  LearnCharacter(u,c) →RecogniseHandwriting*(u,c)* = *in*correct → RecogniseHandwriting*(u,c)* = correct → *Reward(u)* →*NextQuestion(u, c)*
- **Correct on third attempt**
  LearnCharacter(u,c) →RecogniseHandwriting*(u,c)* = *in*correct → RecogniseHandwriting*(u,c)* = *in*correct → RecogniseHandwriting*(u,c)* = correct → *Reward(u)* →*NextQuestion(u, c)*

9.0 Type of Agent

Our proposed application, **Rydit: Learn Mandarin** is a model-based reflex agent, where the model tracks its previous states. The agent tracks the accuracy of the past attempts to decide if a new character shall be introduced and the frequency of the character's appearance. The current percepts will be used to determine the accuracy. The current percepts along with the previous states, allow the agent to make more informed decisions about its actions.

10.0 PEAS Model

The table below shows the formulated solution by using the PEAS model.

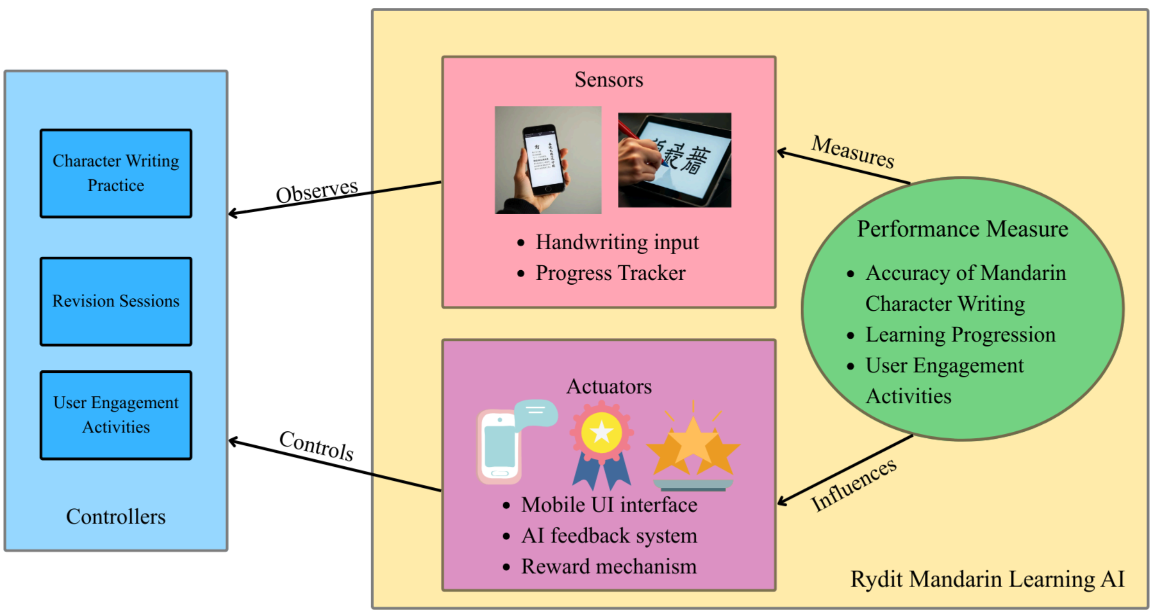
P: Performance Measure	Accuracy of Mandarin character writing, and ● Improvement in user handwriting accuracy (stroke order, direction, and similarity) ● Reduction in user error rate over time ● Successful mastery of characters (≥ 90% accuracy in last 5 attempts) ● Effective retention of characters through spaced repetition Learning progression ● Timely and appropriate introduction of new characters User engagement ● Consistent user engagement and learning streaks ● Positive user feedback and motivation levels (reward points, badges, progress)
E: Environment	Mobile application platform ● Mobile learning environment ● Touchscreen device (tablet or smartphone) ● Gamified elements such as rewards, points, and achievements Learners ● Mandarin learners with varying skill levels Learning sessions ● Learning sessions consisting of: ○ Character revision ○ New word introduction ○ Handwriting practice Mandarin learning context ● Dynamic learning context influenced by: ○ User accuracy ○ Error rate ○ Revision history ○ Learning frequency
A: Actuators/ Effectors	Mobile application interface ● Displaying characters for revision or learning ● Introducing new Mandarin characters ● Moving the user to the next learning question AI feedback system ● Providing real-time handwriting feedback ● Highlighting stroke order and direction corrections ● Adjusting revision frequency using spaced repetition ● Delivering personalised AI feedback (hints, corrections, encouragement) Reward mechanism ● Awarding reward points, badges, and achievements ● Triggering motivational messages and progress notifications
S: Sensor	Handwriting input data

	● User handwriting input (stroke order, direction, shape) User performance data ● User answer correctness ● Accuracy scores from recent attempts ● Error rate per character ● Revision interval data Learning progress indicators ● Learning progress and mastery status ● Frequency and consistency of user practice ● User interaction patterns within the app

11.0 Relations Between Each Property

This diagram shows how the Rydit Mandarin learning app works:

- **Controllers** decide what activity to give next, such as writing practice or revision
- **Sensors** to observe users write and track their learning progress
- **Actuators** respond by giving feedback or rewards
- **Performance measures** to evaluate accuracy, progression and engagement.



12.0 Explanation In Proof Of Concept (POC)

12.1 Performance Measure

Our system is able to help users learn and memorize Mandarin characters through **smart rotation of questions** based on users' exercise patterns. When users are learning to write a character, the system can accurately identify errors when **evaluating user handwriting** based on stroke order, stroke direction and character similarity. The system is able to identify users' weak points and help them overcome even the hardest characters by providing **personalized AI feedback**. The system made learning mandarin painless by gradually increasing the vocabulary, through the **introduction of new words** while ensuring practices on every previously learnt word once in a while. The system can retain users' attention, maintain user engagement and interest in learning Mandarin by giving **rewards** upon goal completion.

12.2 Environment

The operating environment of the system consists of users ranging from 3 to 80 years old that interact with the system using mobile devices. The system operates with a mobile learning environment that uses touchscreen-based handwriting input for learning and practicing Mandarin characters. The environment provides structured learning methods including handwriting practice, character revision and the introduction of new vocabulary. The environment is dynamic and evolves based on user interaction patterns, learning frequency and historical performance data. The system integrates with game-based learning elements, such as reward mechanisms and achievement indicators, which guide the user's learning context and encourages interaction.

12.3 Actuators

The system operates through a mobile application. The mobile interface displays the characters for revision and new characters for learning. The mobile interface will guide the user through each task. Additionally, the AI feedback system provides real-time feedback on handwriting accuracy, including stroke order and direction corrections. After each learning cycle, the system delivers motivational messages and advances the user to the next character. Along with a reward mechanism that awards points, badges, and achievements when positive learning behaviour is detected.

12.4 Sensors

The primary sensor captures the user's handwriting input, including stroke order, stroke direction, and stroke shape. Besides, the system keeps track of the user performance data, including its accuracy and revision intervals for each character. In addition, the system tracks users' learning progress, including the mastery status and practice consistency.

13.0 Behaviours of Agent to Achieve the Goals

In our proposed **Rydit: Learn Mandarin**, the AI agent behaves as a model-based reflex agent. It continuously observes user actions, updates its internal model and takes appropriate action to assist the users in learning Mandarin and achieve their learning goals. Several agent behaviour are implemented in our solution:

1. Performance Monitoring Behaviour
   The agent perceives user handwriting input through sensors, including the handwriting's stroke order, shape and direction. It also records user's recent attempts to extract their error rates and accuracy scores. This is to understand the user's learning progress and maintain an updated internal state in order to provide the most appropriate learning material for the user.

2. Error Detection and Feedback Behaviour
   When incorrect handwriting is detected, the agent provides real-time feedback by highlighting the mistakes and offering corrective guidance for the user. The personalized feedback can help the users to know their errors and improve writing accuracy for next attempts.

3. Adaptive Content Selection Behaviour
   By referencing the stored learning history, the agent decides to introduce new Mandarin characters when mastery is detected and repeat the characters if the error rate is high. The agent controls and selects the learning content that most suits the users to ensure a balanced learning pace and improve users' language skill effectively.

4. Motivation and Engagement Behaviour
   To encourage continuous learning and engagement, the agent triggers awards, badges and positive feedback if the users achieve certain milestones while learning. This includes maintaining a learning streak, showing progress, solving difficulties and reaching a high accuracy when learning.

14.0 Proof of Concept (POC)

Figma Link:
https://www.figma.com/proto/QwHzYNJMFaIKcd8gcnHZFo/AI-Learn-Mandarin-System?node-id=1684-5849&t=NH8VcChS0VTWH1uG-0&scaling=scale-down&content-scaling=fixed&page-id=1114%3A67&starting-point-node-id=1684%3A5849&show-proto-sidebar=1

Video Presentation:
https://youtu.be/OKag_zrU4xg

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